targeted evaluation of reasoning capabilities. Implemented the MATH, pattern induction, and synthetic reasoning scenarios.
Yifan Mai (Infrastructure): Built core infrastructure needed for the short-term and long-term maintenance of this effort. Contributed to the website visualization. Wrote documentation for the codebase.
Yuhui Zhang (Knowledge): Implemented two key adaptation methods for multiple choice tasks. Also implemented the HellaSwag and OpenBookQA scenarios and contributed to the WikiFact scenario.
Yuta Koreeda (Robustness): Helped with implementation of the perturbations for evaluating robustness.

B Core scenarios

B.1 Question answering

B.1.1 BoolQ

Scenario Description. **BoolQ** (Clark et al., 2019) is a question answering scenario that features naturally-occuring yes/no questions. As an example, an input for the scenario looks like: "Elmendorf Air Force Base (IATA: EDF, ICAO: PAED, FAA LID: EDF) is a United States military facility in Anchorage, the largest city in Alaska. Originally known as Elmendorf Field, it became Elmendorf Air Force Base after World War II, and in 2010 it merged with nearby Fort Richardson to form Joint Base Elmendorf-Richardson. Question: Is there an air force base in anchorage alaska?", with the desired output: "Yes". In general, for this scenario, inputs will be a passage followed by a yes/no question, and outputs will be yes/no. Accuracy for this scenario is measured using whether the model generated the correct output, i.e. "Yes" or "No".

Data. Clark et al. (2019) created the dataset by obtaining questions from queries to the Google search engine and heuristically filtering for yes/no questions. Next, human annotators provided answers using Wikipedia articles. The train-dev-test splits for the dataset are 9427-3270-3245 samples.

Pre-processing. Apart from downloading the passage/question/answer triplets from `https://github.com/google-research-datasets/boolean-questions` and tokenizing the instances, no special pre-processing is performed for this dataset. In the few-shot learning setup, due to the limitations of the context length, we limit the number of in-context examples by 5, whereas prior work (Brown et al., 2020) reports using 32 in-context examples.

Data Resources. We access the dataset at `https://github.com/google-research-datasets/boolean-questions`. The dataset is made available through our benchmark. The dataset has no datasheet available.

B.1.2 NewsQA

Scenario Description. **NewsQA** (Trischler et al., 2017) is a question answering scenario that features factual questions related to a given news article. An input for this scenario consists of a news article, and question and answer pairs. For example, a news article "Article: NEW DELHI, India (CNN) – A high court in northern India on Friday acquitted a wealthy businessman...", accompanied with the question "Who was acquitted?" and the associated answer "A wealthy businessman."

In general, for this scenario, inputs will be news articles from CNN along with questions written by crowd-workers related to the article, and the and outputs answers to the questions from the span of the provided news article. Performance for this scenario is measured as the best F1-score of the model output with respect to the reference answers.

Data. Trischler et al. (2017) created this reading comprehension dataset that comprises of 12,744 news articles (90% train, 5% dev, 5% test) and 119,633 question-answer pairs (92549 train, 5166 dev and 5126 test). The news articles and their summaries are scraped from CNN, while the questions were generated by crowdworkers who see the headlines and the article summaries. The answers to these questions were

generated by crowdworkers who saw the entire article. These crowdworkers either select a span of the article as the answer or say that there is "No Answer". A random subset of these question-answer pairs were then validated.

On average, the inputs (article and question) are 856.47 tokens and outputs (reference answers) on average are 2.12 tokens based on the GPT-2 tokenizer.

Pre-processing. We downloaded the dataset by following the instructions at `https://github.com/Maluuba/newsqa`. We process the dataset to only keep news articles where there is at least one valid question. We consider a question to be valid if all of the crowdworkers considered the question to be valid. For each valid question, all of the answers given by crowdworkers, including 'No Answer', are considered to be correct reference outputs.

Data Resources. We accessed the dataset at `https://github.com/Maluuba/newsqa`. The dataset is not made available through our benchmark since each user needs to download the dataset themselves due to legal reasons. The dataset has no datasheet available.

B.1.3 NarrativeQA

Scenario Description. **NarrativeQA** (Kočisky et al., 2017) is a question answering scenario that features long documents and summaries collected from books and movie scripts. Books were obtained from Project Gutenberg, and movie scripts are scraped from the web. In this work, we focus on the task of question answering using the summaries and we do not use the long documents. Given the summary of a long document, the task is to answer non-localized questions. As an example, an input for the scenario looks like: "Mark Hunter (Slater), a high school student in a sleepy suburb of Phoenix, Arizona, starts an FM pirate radio station that broadcasts from the basement of his parents' house. Mark is a loner, an outsider, whose only outlet for his teenage angst [*omitted for brevity*]... Question: Who is Mark Hunter?", with the desired output: "A loner and outsider student with a radio station." or "He is a high school student in Phoenix.". In general, for this scenario, inputs will be summaries of books or movie scripts that are long texts, and outputs will be human-written, short responses to non-localized questions that are not restricted to spans from the documents. Accuracy for this scenario is measured using the F1 score.

Data. Kočisky et al. (2017) created the dataset by collecting passages from books and movie scripts, where books were obtained from Project Gutenberg and movie scripts were scraped from the web. The dataset has 1,567 stories (1,102 training, 115 dev, 355 test). Extracted from these stories, the train-dev-test splits for the dataset have 32,747-3,461-10,557 question-answer pairs.

Pre-processing. To process the dataset, we download the summaries and question/answer pairs from `https://github.com/deepmind/narrativeqa`. Since there are multiple questions per document and testing every question is prohibitively expensive for long-range reasoning tasks, we randomly sample one question per document. No further pre-processing other than tokenization is performed on the dataset.

Data Resources. We access the dataset at `https://github.com/deepmind/narrativeqa`. The dataset is made available through our benchmark. The dataset has no datasheet available.

B.1.4 NaturalQA

Scenario Description. **NaturalQuestions** (Kwiatkowski et al., 2019) is a question answering scenario that features factual naturally-occurring questions. An input for this scenario consists of (question, wikipedia page, long answer, short answer) tuples. For instance, the question "What color was john wilkes booth's hair?", is accompanied by the "`John_Wilkes_Booth`" Wikipedia page, as well as one or more annotator-provided long answers, e.g. "[*omitted for brevity*]...He stood 5 feet 8 inches (1.73 m) tall, had jet-black hair, and was lean and athletic...". Further, for each long answer, a short answer is also provided which most directly answers the question, e.g.: "jet-black" for the example above.

In this work, we focus on the task of predicting the short answer(s) for a given question in both closed- and open-book settings. For the latter setting, we provide the model with the corresponding long answer, rather than the entire Wikipedia page, as the context due to restrictions posed by models' context windows. Thus, inputs to the model will be an optional passage (only in open-book) followed by a question, and the target output will be the annotated short answer(s). Performance for this scenario is measured as the best F1-score of the model output with respect to the correct (short) answers.

Data. Kwiatkowski et al. (2019) created the dataset by obtaining questions based on frequent Google search queries with a Wikipedia page in the top-5 search results. They then relied on annotators to: (i) filter unambiguous factual questions, (ii) provide a *long answer*: an HTML bounding box (if any) within the Wikipedia page that answered the question, and (iii) identify a *short answer*: the span within (ii) that answered the question. For the short answer, annotators could also respond with "Yes", "No" or NULL. The train-dev-test splits for the dataset are 307,373-7,830-7,842, with single and five-way annotations for the train and dev/test splits respectively.

Pre-processing. We download the question/long answer/short answer triplets from `https://storage.googleapis.com/natural_questions/v1.0/` using the dev set exclusively for all our analysis. As described above, in the open-book setting, we provide the long answer as the context to the model. Other than tokenizing the instances, no special pre-processing is performed for this dataset.

Data Resources. We access the dataset at `https://storage.googleapis.com/natural_questions/v1.0/`, and is made available through our benchmark. The dataset has no datasheet available.

B.1.5 QuAC

Scenario Description. **QuAC** (Choi et al., 2018) is a question answering scenario which incorporates dialogue-based context. Specifically, instances are generated through a dialogue between a teacher and a student, where the student tries to learn information about a Wikimedia article that only the teacher has access to. As an example, an input for the scenario looks like: "Title: Daffy Duck, Background: <background section>, Section: Origin & History, Passage: <target section>, Question: What is the origin of Daffy Duck? Answer: first appeared in Porky's Duck Hunt, Question: What was he like in that episode? Answer:" with the desired output: "assertive, unrestrained, combative". In general, for this scenario, inputs will consist of (a) the title of the article, (b) a background paragraph, (c) the title of the target section, (d) the text of the target section, and (e) a sequence of question-answer pairs, where the desired output is the answer to the last question. Accuracy for this scenario is measured using F1-score.

Data. Choi et al. (2018) created the dataset by employing crowd-workers to engage in the dialogue discussed above. The train-dev-test splits for the dataset are 83567 questions for training, 7354 for development, and 7353 for testing.

Pre-processing. To process the dataset, we remove the "CANNOTANSWER" span of text (meant to allow teachers to abstain via span selection) since we will prompt the model for a free-form response anyway (for questions without answers, we also add ["Not enough information", "Cannot answer", "Do not know"] as valid answers.

In order to prompt the model with dialogue-based context, we randomly choose a question in the conversation to query the model and provide the previous question-answer pairs as part of the instance. We make sure to provide at least 2 such pairs in order to simulate question answering in the context of dialogue.

Data Resources. We access the dataset at `http://quac.ai/` and we also make it available through our benchmark. The dataset has a datasheet available at `http://quac.ai/datasheet.pdf`. Further analyses of the dataset were provided by Yatskar (2019).